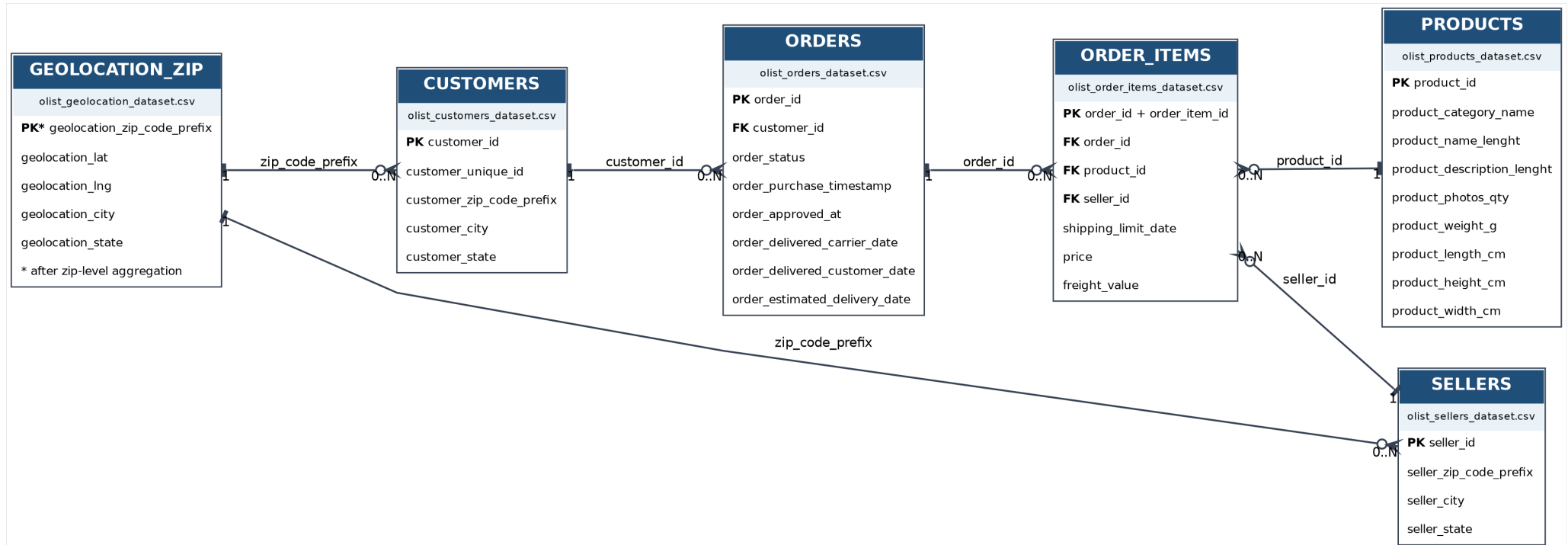


Olist E-Commerce Dataset: ER Model and Data Model

Files covered: orders, customers, order items, sellers, products, and geolocation.

Entity Relationship Model



Recommended final grain	One row per order item, enriched with order, customer, seller, product, and zip-level geolocation attributes.
Geolocation handling	Aggregate geolocation by geolocation_zip_code_prefix before joining. Direct raw joins can duplicate rows because one zip prefix can appear multiple times.

Relationship Summary

Parent Dataset	Parent Key	Child Dataset	Child Key	Relationship
olist_customers_dataset.csv	customer_id	olist_orders_dataset.csv	customer_id	One customer record can map to zero or many orders. In Olist, customer_unique_id should be used for repeat-customer analysis.
olist_orders_dataset.csv	order_id	olist_order_items_dataset.csv	order_id	One order can contain zero or many order item rows.
olist_products_dataset.csv	product_id	olist_order_items_dataset.csv	product_id	One product can appear in zero or many order item rows.
olist_sellers_dataset.csv	seller_id	olist_order_items_dataset.csv	seller_id	One seller can sell zero or many order item rows.
Zip-level geolocation from olist_geolocation_dataset.csv	geolocation_zip_code_prefix	olist_customers_dataset.csv	customer_zip_code_prefix	One aggregated geolocation prefix can map to zero or many customer records.
Zip-level geolocation from olist_geolocation_dataset.csv	geolocation_zip_code_prefix	olist_sellers_dataset.csv	seller_zip_code_prefix	One aggregated geolocation prefix can map to zero or many seller records.

Final Dataset Build Logic

Start with **olist_order_items_dataset.csv** as the base table. Join orders using order_id, customers using customer_id from orders, sellers using seller_id, products using product_id, and zip-level geolocation twice: once for customer location and once for seller location.

Step	Join
1	order_items LEFT JOIN orders ON order_items.order_id = orders.order_id
2	orders LEFT JOIN customers ON orders.customer_id = customers.customer_id
3	order_items LEFT JOIN sellers ON order_items.seller_id = sellers.seller_id
4	order_items LEFT JOIN products ON order_items.product_id = products.product_id
5	customers LEFT JOIN geolocation_zip ON customer_zip_code_prefix = geolocation_zip_code_prefix
6	sellers LEFT JOIN geolocation_zip ON seller_zip_code_prefix = geolocation_zip_code_prefix

Table-by-Table Data Model

olist_orders_dataset.csv

Entity	orders	Primary Key	order_id
Purpose	Order-level transaction table. Contains order lifecycle timestamps and order status.	Grain	One row per order_id.
Column	Data Type	Key	Description
order_id	string	PK	Unique identifier of the order.
customer_id	string	FK	Links the order to olist_customers_dataset.csv.
order_status	string		Order state such as delivered, shipped, canceled, invoiced, or unavailable.
order_purchase_timestamp	datetime		Timestamp when the customer placed the order.
order_approved_at	datetime		Timestamp when the order/payment was approved.
order_delivered_carrier_date	datetime		Timestamp when the order was handed over to the carrier.
order_delivered_customer_date	datetime		Actual delivery timestamp to customer.
order_estimated_delivery_date	datetime		Estimated/promised delivery date.
Notes	Useful derived fields: purchase_year, purchase_month, approval_delay_hours, actual_delivery_days, delivery_delay_days, and is_late_delivery.		

olist_customers_dataset.csv

Entity	customers	Primary Key	customer_id
Purpose	Customer instance and customer location table.	Grain	One row per customer_id. customer_unique_id should be used to identify repeat customers across orders.
Column	Data Type	Key	Description
customer_id	string	PK	Customer identifier used in the orders table.
customer_unique_id	string	Business key	Identifier for the same real customer across multiple orders.
customer_zip_code_prefix	string	FK*	Customer location zip prefix. Join to aggregated geolocation_zip.
customer_city	string		Customer city.
customer_state	string		Customer state.
Notes	For location enrichment, join customer_zip_code_prefix to geolocation_zip_code_prefix after geolocation aggregation.		

olist_order_items_dataset.csv

Entity	order_items	Primary Key	order_id + order_item_id
Purpose	Line-item fact table. Captures product, seller, price, and freight for each item in an order.	Grain	One row per order_id + order_item_id.

Column	Data Type	Key	Description
order_id	string	PK/FK	Links each item row to olist_orders_dataset.csv.
order_item_id	integer	PK	Item sequence number within the order.
product_id	string	FK	Links the item to olist_products_dataset.csv.
seller_id	string	FK	Links the item to olist_sellers_dataset.csv.
shipping_limit_date	datetime		Seller shipping deadline for the item.
price	float		Item price.
freight_value	float		Shipping/freight value for the item.

Notes	This is the recommended base table for the final consumption dataset because it preserves item-level product, seller, price, and freight details.
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olist_sellers_dataset.csv

Entity	sellers	Primary Key	seller_id
Purpose	Seller master and seller location table.	Grain	One row per seller_id.
Column	Data Type	Key	Description
seller_id	string	PK	Unique identifier of the seller.
seller_zip_code_prefix	string	FK*	Seller location zip prefix. Join to aggregated geolocation_zip.
seller_city	string		Seller city.
seller_state	string		Seller state.
Notes	For location enrichment, join seller_zip_code_prefix to geolocation_zip_code_prefix after geolocation aggregation.		

olist_products_dataset.csv

Entity	products	Primary Key	product_id
Purpose	Product master table with category and physical dimensions.	Grain	One row per product_id.
Column	Data Type	Key	Description
product_id	string	PK	Unique product identifier.
product_category_name	string		Product category name.
product_name_lenght	integer		Length of the product name. Original dataset uses spelling "lenght".
product_description_lenght	integer		Length of the product description. Original dataset uses spelling "lenght".
product_photos_qty	integer		Number of product photos.
product_weight_g	float		Product weight in grams.
product_length_cm	float		Product length in centimeters.
product_height_cm	float		Product height in centimeters.
product_width_cm	float		Product width in centimeters.
Notes	Useful derived fields: product_weight_kg and product_volume_cm3.		

olist_geolocation_dataset.csv

Entity	geolocation	Primary Key	No strict raw primary key. Use geolocation_zip_code_prefix after aggregation.
Purpose	Zip-prefix-level latitude, longitude, city, and state source table.	Grain	Raw file can contain multiple rows per geolocation_zip_code_prefix. Use an aggregated zip-level version for joins.
Column	Data Type	Key	Description
geolocation_zip_code_prefix	string	Business key*	Zip prefix used to map customers and sellers to latitude/longitude.
geolocation_lat	float		Latitude.
geolocation_lng	float		Longitude.
geolocation_city	string		City for the zip prefix.
geolocation_state	string		State for the zip prefix.
Notes	Recommended aggregation: group by geolocation_zip_code_prefix and take mean latitude/longitude. Resolve city/state using the most frequent value per zip prefix.		

Recommended Final Consumption Dataset

Recommended final table name: **fact_order_items_enriched**

Grain: one row per order item.

Group	Recommended Columns
Identifiers	order_id, order_item_id, customer_id, customer_unique_id, product_id, seller_id
Order Status and Dates	order_status, order_purchase_timestamp, order_approved_at, order_delivered_carrier_date, order_delivered_customer_date, order_estimated_delivery_date
Commercial Values	price, freight_value, item_total_value, freight_percentage
Product Attributes	product_category_name, product_name_lenght, product_description_lenght, product_photos_qty, product_weight_g, product_weight_kg, product_length_cm, product_height_cm, product_width_cm, product_volume_cm3
Customer Location	customer_zip_code_prefix, customer_city, customer_state, customer_lat, customer_lng
Seller Location	seller_zip_code_prefix, seller_city, seller_state, seller_lat, seller_lng
Derived Features	purchase_year, purchase_month, purchase_day, purchase_dayofweek, purchase_hour, seller_customer_same_state, seller_customer_same_city, approval_delay_hours, carrier_handover_days, actual_delivery_days, estimated_delivery_days, delivery_delay_days, is_late_delivery

Data Preparation Rules

Rule	Recommendation
Date fields	Parse all order lifecycle columns and shipping_limit_date as datetime.
Zip prefixes	Convert zip prefix columns to strings and left-pad to 5 digits where needed.
Geolocation	Aggregate by zip prefix before joining to avoid row multiplication.
Product spelling	Keep raw columns product_name_lenght and product_description_lenght as-is, or rename only after ingestion.
Missing categories	Fill missing product_category_name with unknown for analytics/ML usage.
Final grain	Do not aggregate order_items unless the use case specifically requires order-level data.